Implementing Privacy-Preserving Techniques for Secure Healthcare Data Analytics

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Objectives

- To protect sensitive healthcare data using strong encryption techniques
- To ensure patient data privacy during data storage and data analysis
- To implement secure data analytics using privacy-preserving methods
- To apply machine learning techniques on encrypted healthcare data
- To prevent unauthorized access and avoid data leakage

Existing Systems

- Healthcare data is stored in centralized databases
- Most systems use plain or weakly protected data storage
- Patient data is directly accessed during analysis
- Limited privacy protection mechanisms are available
- High risk of data leakage and unauthorized access

Proposed System

- AES encryption is used to secure healthcare data by converting sensitive patient information into encrypted form.
- Healthcare data is encrypted before storage and processing to avoid exposure of raw patient details.
- Patient privacy and data confidentiality are ensured throughout the data handling process.
- Machine learning techniques are applied on secure data for healthcare analysis and prediction.
- Unauthorized access and data leakage are prevented using encryption and access control mechanisms.

System Requirements

Software Requirements

- Python
- Google Colab / Jupyter Notebook
- Machine Learning Libraries
  - NumPy
  - Pandas
  - Scikit-learn
  - Matplotlib / Seaborn
  - PyCryptodome (for AES encryption)

Literature Survey

S.No	TITLE	AUTHOR(S)	INFERENCE	PUBLISHED IN
1.	Privacy Preservation In Artificial Intelligence - Enabled Healthcare Analytics	Shancang Li1* , Muddesar Iqbal2 , Ali Kashif Bashir3,5,6 and Xinheng Wang4	privacy-preserving methods improve secure data storage and controlled access in healthcare systems.	RESEARCH (Research Manuscript) Open Access
2.	Privacy - Preserving AI For Secure Data Sharing In Healthcare	Rajashekhar Reddy Kethireddy	encryption-based privacy techniques can securely enable healthcare data analysis without compromising patient confidentiality	Journal of Recent Trends in Computer Science and Engineering (JRTCSE)
3.	AI- Driven Secure Data Sharing: A Trustworthy And Privacy - Preserving Approach	Al Amin ⋆ , Kamrul Hasan ⋆ , Sharif Ullah† , Liang Hong ⋆ ⋆ Tennessee State University, TN, USA	encrypted data can be effectively used in advanced machine learning models while maintaining high privacy and accuracy.	University of Central Arkansas, Conway, AR, USA
4.	Privacy-Preserving Machine Learning For Healthcare: Open Challenges And Future Perspectives	Alejandro Guerra-Manzanares ⋆ , L. Julian Lechuga Lopez ⋆ , Michail Maniatakos and Farah E. Shamout	combining security mechanisms with machine learning enhances reliable and safe healthcare predictions.	ICLR 2023 Workshop on Trustworthy Machine Learning for Healthcare

Architecture Diagram:

Start

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Patient / Healthcare Data Collection

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Preprocessing of Healthcare Data

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AES Encryption of Sensitive Data

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Secure Storage of Encrypted Data

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Authorized User Authentication

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Decryption Using Secret Key

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Machine Learning Model (Logistic Regression)

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Disease Prediction / Data Analysis

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Secure Result Display to Authorized User

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End

Modules

- Data Acquisition Module
  - Collects and preprocesses sensitive dataset for analysis.
- AES Encryption Module
  - Encrypts the dataset to ensure data confidentiality.
- Secure Storage Module
  - Stores the encrypted data safely in the system.
- Authentication & Decryption Module
  - Allows only authorized users to decrypt data using a secret key.
- Machine Learning Module
  - Applies Logistic Regression for classification and prediction.
- Result & Visualization Module
  - Displays prediction results with graphical representation.

Work Plan & Timeline

Phase	Activities	Duration	Phase	Activities
Phase 1	Problem Definition & Literature Survey	2 weeks	Phase 1	To securely analyze healthcare data using machine learning while preventing privacy breaches and unauthorized access
Phase 2	Design & Implementation	4 weeks	Phase 2	Data masking,Anomaly detection ,Noise addition
Phase 3	Techniques Used	4 weeks	Phase 3	AES technique is used in this part and by that we are encrypting the datas
Phase 4	Model Training & Evaluation	1 weeks	Phase 4	Machine learning model like random forest technique is used

THANK YOU